

Southwest Heat Wave

August 2020 · rung 1 of 9

observed A_L

+5.14 K

ERA5, verification week

deepest walker

+4.31 K

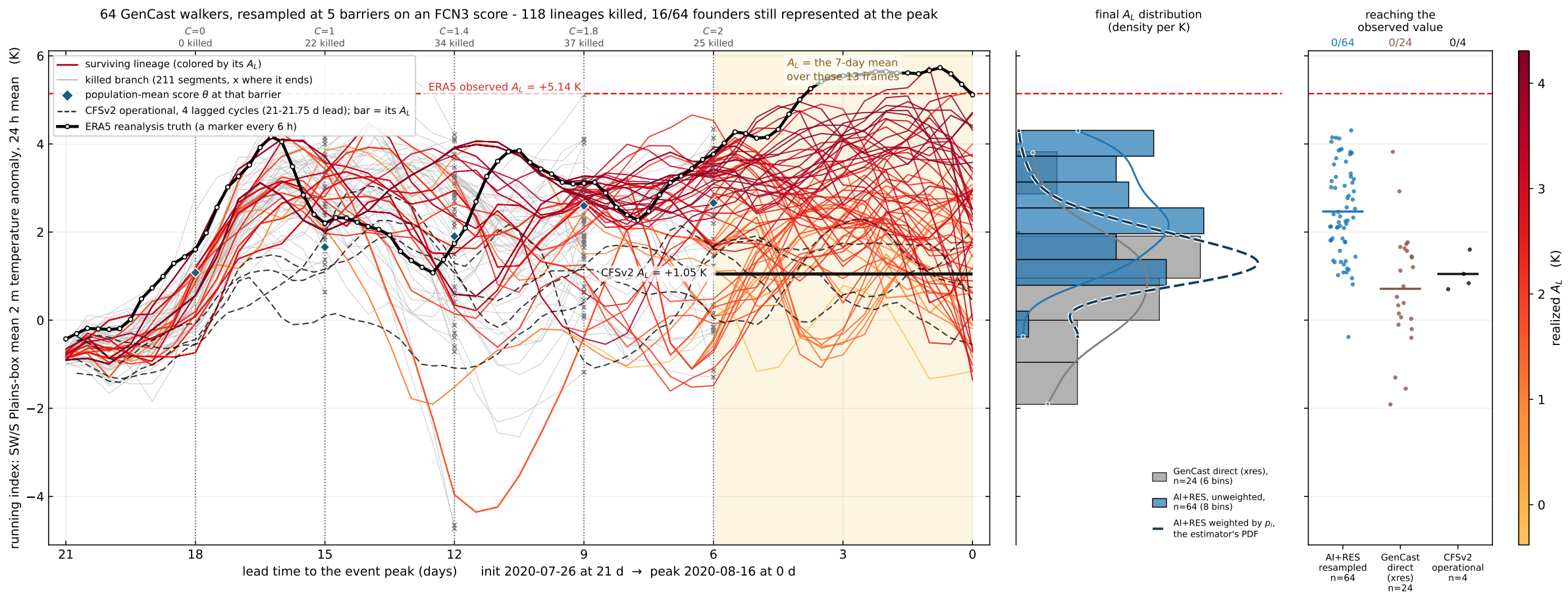
0 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

$< 3 \times 10^{-4}$

sum of the walker probabilities

AI+RES pilot - Southwest_HeatWave_2020 (pilot): the walk itself, barrier by barrier



California Heat Wave

September 2022 · rung 2 of 9

observed A_L

+4.71 K

ERA5, verification week

deepest walker

+6.33 K

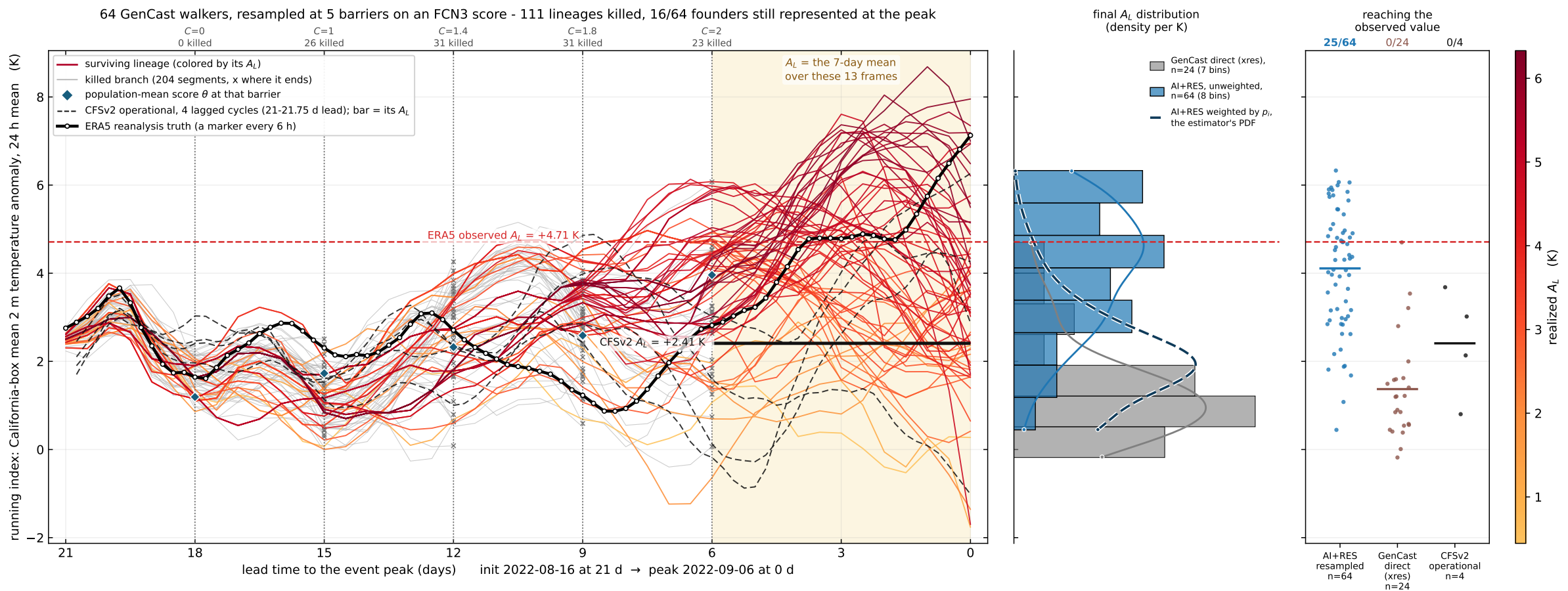
25 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

0.0186

sum of the walker probabilities

AI+RES pilot - California_HeatWave_2022 (pilot): the walk itself, barrier by barrier



Winter Storm Elliott

December 2022 · rung 3 of 9

observed A_L

-14.42 K

ERA5, verification week

deepest walker

-17.04 K

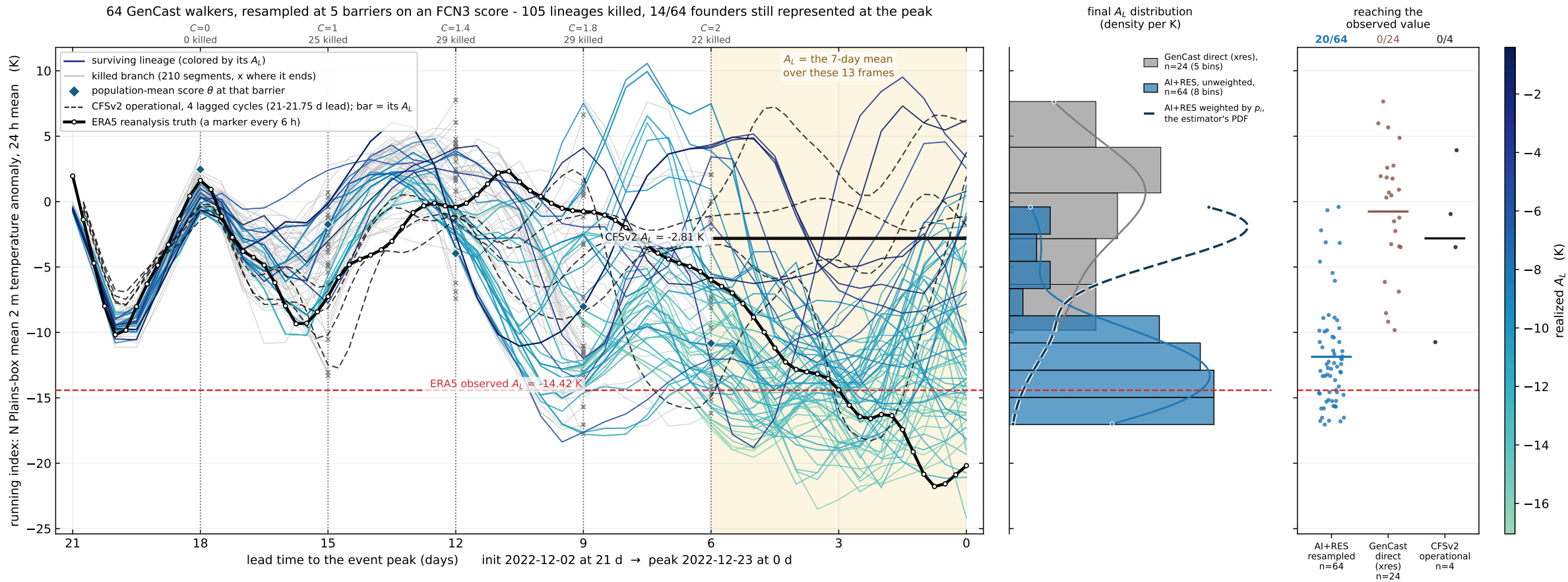
20 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

0.0079

sum of the walker probabilities

AI+RES pilot - WinterStorm_Elliott_2022 (pilot): the walk itself, barrier by barrier



Winter Storm Uri

February 2021 · rung 4 of 9

observed A_L

-7.40 K

ERA5, verification week

deepest walker

-10.20 K

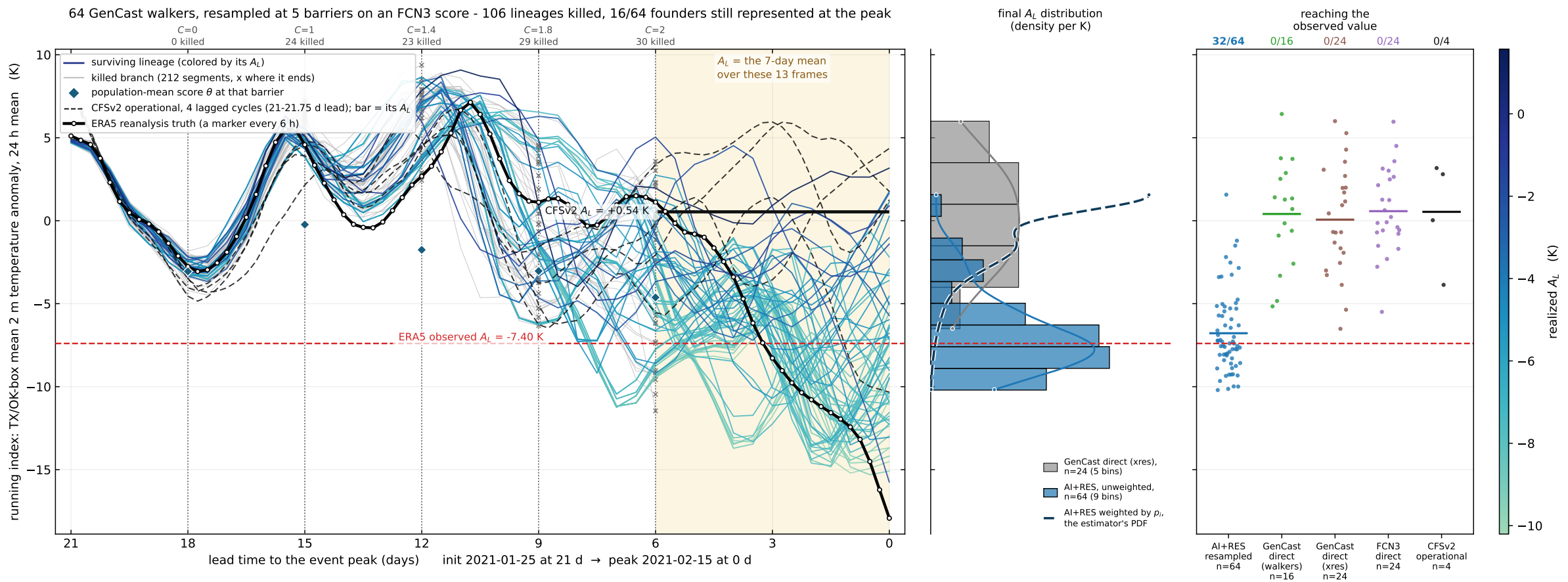
32 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

0.0090

sum of the walker probabilities

AI+RES pilot - WinterStorm Uri_2021 (pilot): the walk itself, barrier by barrier



South-Central Heat Dome

June 2023 · rung 5 of 9

observed A_L

+5.73 K

ERA5, verification week

deepest walker

+6.19 K

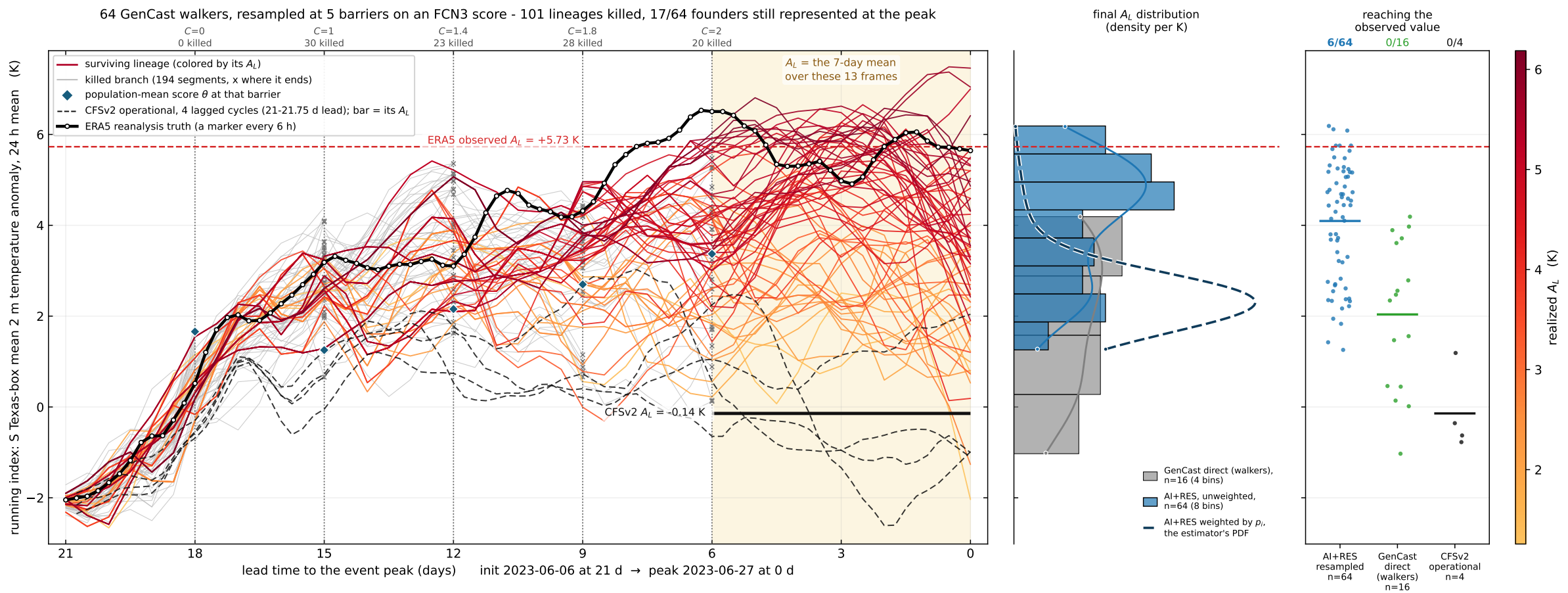
6 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

0.00273

sum of the walker probabilities

AI+RES pilot - SCentral_HeatDome_2023 (pilot): the walk itself, barrier by barrier



Pacific Northwest Heat Dome

June 2021 · rung 6 of 9

observed A_L

+7.72 K

ERA5, verification week

deepest walker

+10.29 K

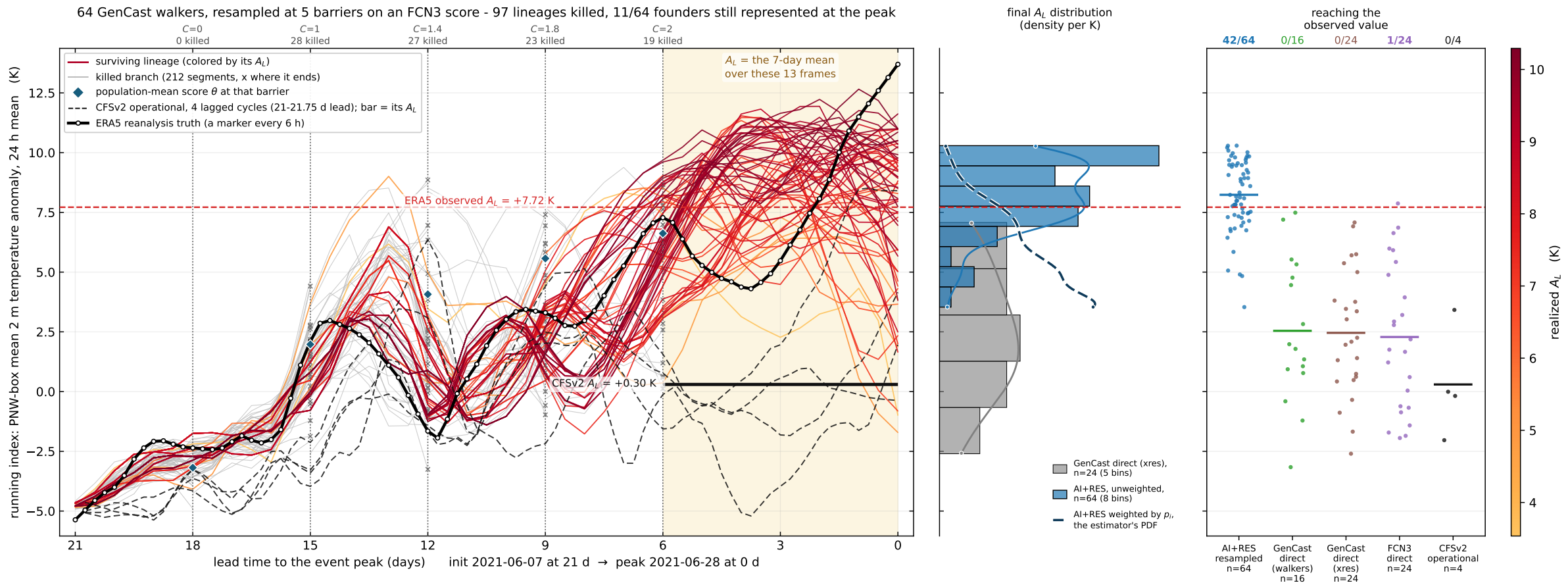
42 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

0.054

sum of the walker probabilities

AI+RES pilot - PNW_HeatDome_2021 (pilot): the walk itself, barrier by barrier



The Nationwide Warm Spell

24 December 2025 · rung 7 of 9

observed A_L

+3.26 K

ERA5, verification week

deepest walker

+5.12 K

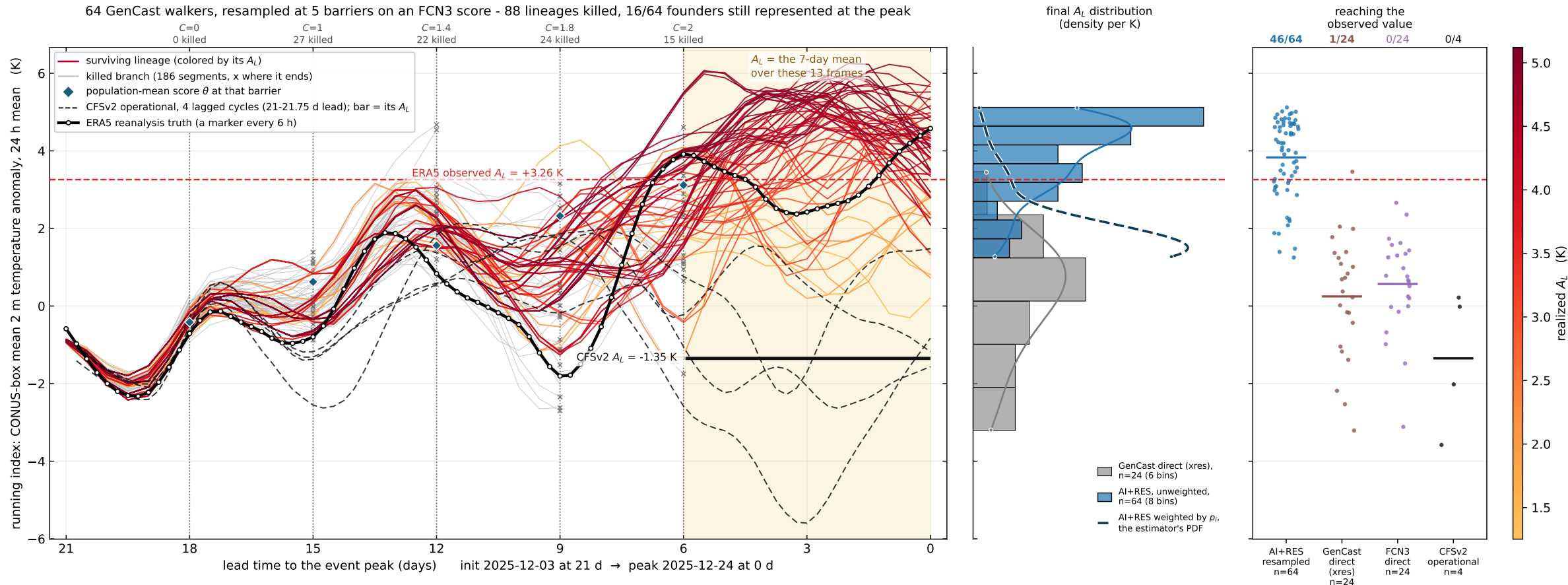
46 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

0.0661

sum of the walker probabilities

AI+RES pilot - p90_20251224 (pilot): the walk itself, barrier by barrier



The Normal-Day Test

2 August 2024 · rung 8 of 9

observed A_L

+1.06 K

ERA5, verification week

deepest walker

+2.37 K

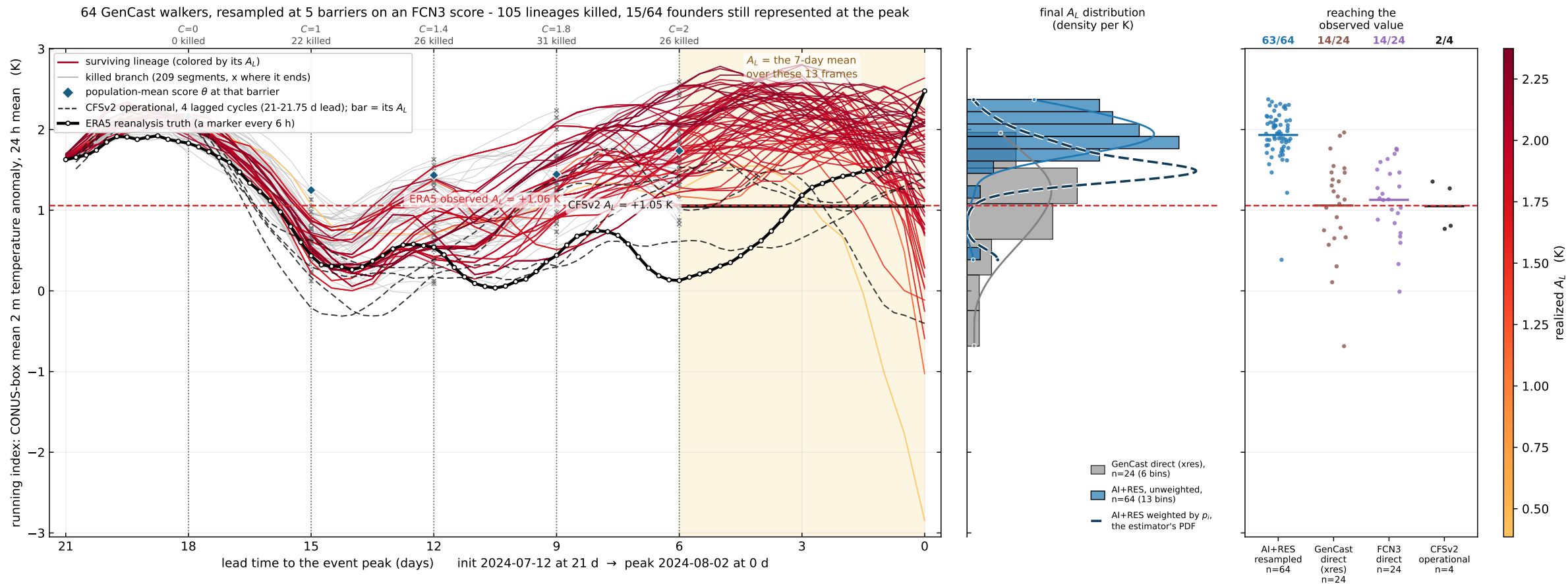
63 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

0.649

sum of the walker probabilities

AI+RES pilot - p90_20240802 (pilot): the walk itself, barrier by barrier



The Below-Median Day

7 November 2023 · rung 9 of 9

observed A_L

+0.21 K

ERA5, verification week

deepest walker

+4.16 K

64 / 64 walkers past the observation

$P(A_L \text{ beyond observed})$

0.583 (pinned)

sum of the walker probabilities

AI+RES pilot - p90_20231107 (pilot): the walk itself, barrier by barrier

